AutoVAG: Automating Visual Attention Guidance in Learning Environments

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Abstract

Grounding spoken lecture content to slide regions provides visual emphasis that boosts student engagement, but automation remains challenging due to poor model generalization, driven by the lack of open training data with text-rich academic content and select-one-of-many grounding tasks. We present AutoVAG, a novel framework that reframes cross-modal grounding as a text-to-text reasoning task. Specifically, AutoVAG uses a Vision-Language Model (VLM) to generate textual descriptions of each Area of Interest (AOI) and formulates AOI selection as a multiple-choice question (MCQ) answered by a fine-tuned Large Language Model (LLM). To enable large-scale training data generation, we introduce a weakly supervised pipeline that leverages natural pointer cues from lecture recordings to automatically produce labeled data, yielding a dataset of 479,000 subtitle–AOI pairs. AutoVAG achieves 52.3% top-1 accuracy on our dataset of 37 courses, outperforming a text-contrastive model (44.7%) and VLM-based grounding approaches (37.0%). Furthermore, AutoVAG generalizes effectively across unseen languages and academic domains and generates selection rationales that strengthen user confidence.

Introduction

A student's visual attention is a key determinant of learning outcomes. Traditionally, instructors use laser pointers, either physical devices or virtual pointers integrated into presentation software, to highlight important regions of slides and to quickly direct attention to critical content within dense materials, leading to significantly enhanced comprehension and retention (Sauter, Wagner, and Huckauf 2022; Sharma, Giannakos, and Dillenbourg 2020). However, consistently and effectively using a laser pointer in the classroom can be challenging. Instructors often face a high cognitive load as they operate slide transitions and animations, annotate slides in real time, and explain complex content, making it difficult to maintain precise pointer control. Additionally, some may find the method ineffective or avoid it due to constraints in classroom layout or available equipment.

If *visual attention guidance can be automatically generated* based on an understanding of the course materials and the instructor's speech, the technique of pointer-based attention guidance could be adopted at scale, both in real time and in offline contexts, and across settings such as physical classrooms and video-based learning environments.

In this paper, we introduce AutoVAG, a system that automatically identifies the most relevant regions of a slide based on the instructor's speech. These regions can then be visually emphasized in either projected presentations or recorded videos, supporting student learning in both physical classrooms and online learning platforms.

Despite the importance of the task, there is no established method with satisfactory results, both in the case of conventional methods such as CLIP (Radford et al. 2021), OCR–image fusion pipelines (Lee et al. 2023), document parsers (Li et al. 2022), and the recently surged multimodal Large Language Models (LLMs) (Peng et al. 2023; Bai et al. 2025; Lai et al. 2024). The former fail to gain deep insight in the context beyond the superficial correlation between the text and the image. Meanwhile, the latter often struggle with most lecture slides–text-rich images of documents (Li et al. 2025) with the complex layouts and textual content.

Specifically, there are two associated challenges: **(1) domain knowledge gap**. Current models are mostly trained on natural images and are not equipped to interpret the structured, dense, and domain-specific nature of lecture slides, which often contain charts, mathematical formulas, and code snippets. It remains challenging for existing methods to perform precise localization and understand these content, especially the textual one, and the interactions between multiple objects. **(2) data scarcity**. To the best of our knowledge, no large-scale public dataset aligns spoken lecture content with corresponding visual Areas of Interest (AOIs). Moreover, existing grounding models are trained to only ground a *single* object (Xiao et al. 2024), significantly different from the *select-one-out-of-many* task. Creating such a dataset is essential for model training yet remains non-trivial, as large-scale manual annotation to produce a labelled dataset is expensive and impractical.

To tackle the challenge of the domain knowledge gap, our key idea is to reframe the retrieval task into the text domain. Instead of using general-purpose Visual Language Models (VLMs) (Li et al. 2023), which often struggle to understand individual content from the entire slide image, we adopt a divide-and-conquer strategy: decomposing a slide image into Areas of Interest (AOIs) and querying VLMs to generate semantically rich and accurate textual descriptions of these AOIs. The former facilitates VLMs understanding individual visual content, while the later enables

us to leverage the strong language understanding, contextual reasoning, and inference capabilities of modern LLMs to perform robust text-to-text matching between spoken subtitles and AOI descriptions. Furthermore, operating in the textual space enhances generalization across both languages and academic domains.

To address the issue of data scarcity, we automate the annotation process with a robust and scalable multimodal pipeline, leveraging open tools and models. This allows us to process diverse real-world lecture videos and create weakly labeled data at scale. The key insight is that even slightly inaccurate visual cues, such as laser pointer movements, can contain useful information. Our approach combines efficient laser pointer detection with fine-tuned Document Image Transformer (DIT) for granular AOI segmentation, and a temporal sliding window for context-aware subtitle-AOI alignment. We demonstrate that scalable, domain-robust supervision can be achieved at low cost and the larger amount of training data enabls effective training of the downstream AOI retrieval model in real-world teaching environments.

Our main contributions are as follows. (1) A scalable annotation pipeline that exploits instructor pointer cues and automated slide parsing to produce over 3× more subtitle–AOI pairs than manual methods; (2) The design and thorough evaluation of an LLM-enabled multiple-choice retrieval approach built on VLM-generated AOI descriptions, which outperforms OCR+image-fusion and contrastive-learning baselines; (3) Extensive experiments on 37 real-world EECS courses demonstrating 52.3 % Top-1 accuracy, strong generalization across languages and domains, and practical benefits in efficiency and interpretability; (4) User study over real-world users, validating the effectiveness of the proposed method in real-world use case.

Related Work

Visual Attention and Learning in Educational Settings Guiding visual attention is crucial for effective multimedia learning (Mayer and Moreno 2003), and instructor cues like pointer usage are established methods to direct focus and improve learning outcomes (Pi et al. 2019; Sung and Mayer 2012), aligning with the eye-mind hypothesis (Just and Carpenter 1980). Eye-tracking studies by Sauter et al. (Sauter, Wagner, and Huckauf 2022) and Sharma et al. (Sharma, Giannakos, and Dillenbourg 2020) empirically confirm that students who closely follow an instructor's visual cues (e.g., pointers, deictic acts) tend to achieve better learning gains. While these studies affirm the importance of attending to highlighted areas, their common reliance on controlled settings or manual analysis underscores the need for scalable, automated methods to leverage these attention-learning dynamics in real-world lecture videos.

Datasets for Lecture Video Understanding Progress in lecture video understanding is deeply intertwined with the evolution of supportive datasets, as reviewed by Lee et al. (Lee et al. 2023) and Li et al. (Li et al. 2024). While earlier datasets addressed diverse specific tasks (e.g., (Li et al. 2019; Bulathwela et al. 2020; Dutta et al. 2018; Chen et al. 2014)), the Multimodal Lecture Presentations (MLP) Dataset (Lee et al. 2023) offers a more comprehensive benchmark with aligned multimodal data, widely used for retrieval research (Li et al. 2024). However, the creation of MLP's core structural annotations (e.g., slide segments, figure boundaries) involves significant manual effort. This limitation in scalable, automated generation of fine-grained (spoken subtitle, visual AOI) pairings from raw lecture data using weak signals motivates our development of a highly automated pipeline.

Multimodal and Cross-Modal Retrieval Cross-Modal Retrieval (CMR) for educational content is challenging due to the complexity of lecture slides and the scarcity of educational data. Aside from specialized models like PolyViLT (Lee et al. 2023) and EduCross (Li et al. 2024), VLM-based methods have gained increasing interest, which can be categorized into two camps: (1) end-to-end learning methods that train VLMs (Peng et al. 2023; Bai et al. 2025) to directly detect referred AOIs (Karthik et al. 2023; Lin et al. 2024; Qu et al. 2024) in the image, and (2) multi-stage approaches that train VLMs to perform multiple intermediate tasks before localizing the AOIs. Our AutoVAG falls into the second camp, that decomposes a slide image into AOIs, describes an AOI in texts and considers subtitle-AOI alignment as a text-text matching problem.

Method

To automatically ground spoken lecture content to visual Areas of Interest (AOIs) and enable robust subtitle–AOI retrieval, our approach integrates three core components: (1) a process for transforming visual AOIs into rich semantic descriptions using Visual Language Models (VLMs) and (2) text-based retrieval mechanisms to align subtitles with these AOI descriptions. Moreover, we propose an automated pipeline for generating weakly supervised subtitle-AOI annotations. The proposed annotation pipeline and framework enables us to address challenges of data scarcity and domain specificity inherent in lecture materials. We present an overview of AutoVAG framework in Figure 1.

Automated Weakly Supervised Annotation Pipeline

To overcome data scarcity, we develop an automated pipeline to generate large-scale subtitle–AOI pairs. First, we collect a dataset of orally lectured videos, which are visually presented with slides and guided with lazer pointers by the instructors. Next, we temporally align the instructor's caption with a region of the slide based on the localization cues of pointers. We detail the annotation pipeline as follows:

Video slide segmentation A lecture video is first decomposed into individual slides based on the moments of slide transitions. We automatically detect slide transitions based on visual changes between consecutive video frames. We adopt a combination of perceptual hashing (Zauner 2010) and Structural Similarity Index Measure (Wang et al. 2004) as the metric to measure the visual change. For the entire dataset, we empirically set a threshold to distinguish the occurence of slide transitions from intra-slide visual updates, such as animated figures in a slide.

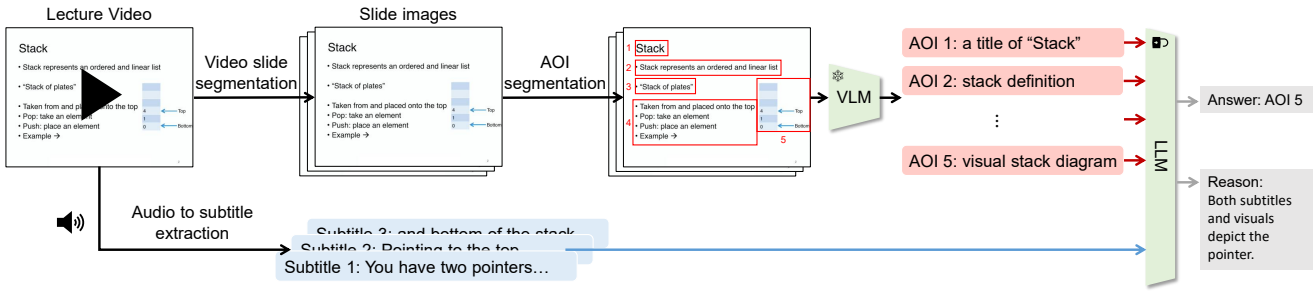


Figure 1: Overview of AutoVAG framework. Given a lecture video, AutoVAG localizes a spoken subtitle on the slide image in three stages: (1) segmenting the slide image into AOIs, (2) describing each AOI in texts with VLMs, and (3) applying a fine-tuned LLM to align the input subtitle with the text of the referred AOI. In addition to predicting the referred AOI index, the LLM of AutoVAG is able to generate reasoning for its prediction.

AOI segmentation A slide image often contains different contents in form of local regions on the image. We pick out these contents by detecting the corresponding Area of Interest (AOI). In this paper, we define six types of AOI–textttimage, `math`, `plot`, `table`, `text`, and `title`–for educational lecture. Specifically, we fine-tune a Document Image Transformer with an RCNN head (Li et al. 2022; Girshick 2015) on a set of manually labelled lecture slides to accurately identify AOIs.

Grounding subtitle Lecture audio is transcribed into timestamped subtitle segments using COOL-whisper (Tseng et al. 2024). To provide more context, we expanded each subtitle segment by including the subtitles from 15 seconds before and after the original segment. For each subtitle segment, the video timestamp and the corresponding slide image can be localized based on the temporal midpoint t_{mid}.

To pinpoint the instructor's visual focus for each subtitle, we spatio-temporally associate the spoken subtitle with a region on the slide using the on-screen pointer coordinates. We automatically detect the pointer in a slide with a YOLOv8 (Varghese and M. 2024) object detector and localize the subtitle with detected pointer coordinates in the slide at t_{mid}. The YOLOv8 detector is fine-tuned on a diverse synthetic dataset of pointer instances (cursors and laser dots) overlaid on various slide backgrounds with augmentations, to enhance detection capabilities.

With the pointer coordinate at t_{mid} of a a subtitle segment, we can spatially match the subtitle with the nearest AOI on the image. Collectively, the fully automated pipeline of video slide segmentation, AOI segmentation and grounding subtitle enable us to generate a large-scale dataset, containing $3\times$ more subtitle–AOI pairs than MLP dataset (Lee et al. 2023), with minimal human effort.

AutoVAG

Semantic descriptions of AOI AOIs in lecture slides are often rich in semantic content, including mathematical formulas, tables, plots, and animated figures. This diversity of visual content makes it challenging for a model to learn the alignment between visual AOIs and textual subtitles. It must jointly learn to understand these visual elements and their corresponding cross-modal alignment with the subtitles.

To bridge the gap between complex visual AOIs and textual subtitles, we propose to convert each segmented AOI image into a natural-language description. As a result, the AOI-subtitle alignment problem can be simplified into a text-to-text alignment problem.

We use Pixtral-12B (Agrawal et al. 2024) to extract textual description from each AOI image. However, the model has limited understanding in Mandarin. For AOIs with Mandarin content, we deploy a fine-tuned Phi-4 Multimodal model (Abouelenin et al. 2025) instead. These VLMs are prompted to generate informative summaries capturing both semantic content and visual structure of the AOI.

Since most lecture videos in our dataset are narrated in Mandarin, we additionally translate the text descriptions of AOIs into Mandarin using LLaMA 3.1 8B (Grattafiori et al. 2024). However, our method is general to any other language. We refer to the Appendix for more details.

Text-to-Text Alignment To tackle the text-to-text alignment problem, we investigate two methods: contrastive learning and LLM-based multi-choice question answering.

For contrastive learning, both the subtitle and the textual descriptions of all AOIs are encoded into latent features using the text encoder of CLIP (Radford et al. 2021). We use a Mandarin version of CLIP–Chinese-CLIP (Yang et al. 2022)–to process our dataset. The model is trained with a cross-entropy loss to maximize the similarity between a subtitle and its paired AOI text description, while simultaneously minimizing its similarity to other AOI descriptions from the same slide. This encourages the learning of a joint embedding space where related text pairs are close.

For LLM-based multi-choice question answering, a LLM is prompted to select the ground-truth AOI from all AOIs on the same slide. The model takes as input a question (subtitle) and multiple numbered options (textual descriptions of AOIs); it outputs the index of the selected answer. The model is trained with next-token-prediction loss using LoRA(Hu

et al. 2022). Compared to contrastive learning, this method has an advantage of leveraging the LLM's extensive commonsense knowledge and reasoning capabilities for robust matching.

Explainability of subtitle-to-AOI alignment The LLM-based method has another advantage over contrastive learning methods–generating reasoning texts to enhance the explainability of the decision making progress. We fine-tune the LLM, pre-trained for multi-choice question answering, to generate human-readable rationales for its predictions. In particular, we deploy an LLaMA 3.1 8B–to reason the association between a subtitle segment and the text description of a AOI. We use the reasoning results as pseudo ground-truth and fine-tune our LLM to generate both the AOI index and the corresponding reason. During inference, the LLM is iteratively prompted to predict the aligned AOI index and then the reason for its prediction.

Experiments

Dataset

In-domain dataset We curated a dataset comprising 37 undergraduate-level electrical engineering and computer science (EECS) courses. To the best of our knowledge, *this is the largest visual grounding dataset in the educational domain*. It contains 30,396 slides—three times larger than the MLP dataset (Lee et al. 2023). As shown in Table 1, six courses were taught entirely in English, while the remaining 31 were delivered primarily in Mandarin with English support. We split the dataset into a training set (26 Mandarin courses, 3 English courses) and a validation set (5 Mandarin courses, 3 English courses). All lectures included both slides and oral presentations.

We pre-processed the dataset using our automated annotation pipeline described earlier. As summarized in Table 1 and illustrated in Figure 2, across the 37 courses we detected 138,362 distinct AOIs and 479,416 subtitle–AOI pairs. Among these AOIs, 49.9% correspond to text, 21.% to images, 15.4% to titles, 11.1% to math, 1.5% to plots, and 1.1% to tables. The high diversity of AOI categories poses significant challenges for accurate subtitle–AOI pairing.

Out-of-domain dataset As shown in Table 1, we additionally curated two datasets to evaluate the multilingual and multidisciplinary generalization of AutoVAG. For the novel-language dataset, we collected two courses[1] from another university's open online course platform, both of which were taught in Japanese. For the novel-content dataset, we collected four courses[2] covering subjects in management. Both datasets were pre-processed using the same automated annotation pipeline described earlier.

Due to institutional policies and privacy considerations, we are unable to publicly release the entire dataset. However, we will release a subset of courses that are already available on public platforms such as YouTube to support the reproducibility of our method.

[1] *Infrared Astronomy and Mathematics* and *Topological Data Analysis*

[2] *Microeconomics*, *Macroeconomics*, *Manufacturing Data Science*, and *Big Data and Business Analytics*

Experimental Setup

Baselines **(1) PolyViLT**: The method (Lee et al. 2023) learns the mapping from slide images to subtitles via multi-instance loss. We adapted this method to our subtitle–AOI alignment task by extracting AOI text using EasyOCR (JaidedAI 2020), tokenizing it with BERT (Devlin et al. 2019), and encoding AOI images using the ViLT image encoder. **(2) Contrastive learning**: the method learns AOI-to-subtitle alignment through text-to-text contrastive learning. We encoded both subtitles and AOI descriptions (generated by VLMs) using the Chinese-CLIP text encoder (Yang et al. 2022). **(3) VLM-based visual grounding**: The method uses a VLM to predict the location of the AOI referenced by a subtitle (Peng et al. 2023). The VLM takes the entire slide image and the subtitle as input and outputs the bounding box coordinates of the corresponding AOI. **(4) VLM-based AOI selection**: The method annotates each AOI on a slide with numbered marks (Xuan et al. 2024) and prompts the VLM to select the referenced AOI index from the annotated image.

For baselines (3) and (4), we adopted Qwen 2.5-VL (Bai et al. 2025) as the VLM. For each VLM-based baseline, we evaluated both the pre-trained model and its fine-tuned counterpart using our dataset.

Training and testing We first trained AutoVAG and all baseline models on the training set of Mandarin-lectured EECS courses. The validation set from these Mandarin courses was then used for in-domain evaluation. To assess multilingual generalization, we tested the models on English-lectured EECS courses and Japanese-lectured courses. For multidisciplinary generalization, we evaluated the models on management courses. Finally, we fine-tuned AutoVAG and the contrastive learning baseline on the English-lectured EECS courses to examine whether fine-tuning on a small amount of in-domain data improves performance.

Evaluation metrics We adopted accuracy as the evaluation metric, defined as $\mathrm{Acc.} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}(\hat{y}_i = y_i)$, where N is the number of test samples, and $\hat{y}_i$ and y_i are the predicted and ground-truth AOIs for subtitle i, respectively. For AutoVAG, VLM-based AOI selection, PolyViLT, and the contrastive learning baseline, $\hat{y}_i$ and y_i represent the indices of the predicted and ground-truth AOIs. For VLM-based visual grounding, however, $\hat{y}_i$ and y_i denote the bounding box coordinates of AOIs, and in this case $\hat{y}_i = y_i$ holds if the intersection over union (IoU) between the predicted and ground-truth bounding boxes is greater than or equal to 0.5.

Results

In-domain Mandarin-lectured EECS courses As shown in Table 2, our AutoVAG sets a new state-of-the-art on the validation set of Mandarin-lectured EECS courses. Owing to the strong language understanding capabilities of LLMs, our method outperforms contrastive learning baseline and PoliViLT by a large margin ($+7.6\%$ and $+19\%$). Notably,

Benchmark	#. Courses	Train	Valid.	#. Pages	#. Pairs	#. Images
In-domain dataset						
EECS (Mandarin)	31	26	5	29,222	434,728	27206
EECS (English)	6	3	3	4,998	44,688	2016
Total	**37**	**29**	**8**	**30,396**	**479,416**	**29,222**
Out-of-domain dataset						
Management courses	4	–	4	1,986	23,546	1,113
Japanese courses	2	–	2	105	514	164

Table 1: Statistics for our dataset of subtitle–AOI pairs.

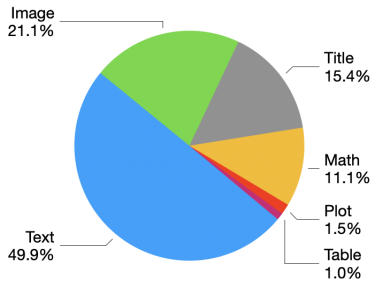
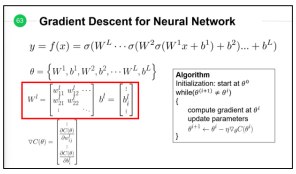


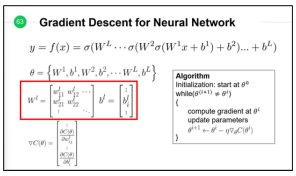
Figure 2: Distribution of AOI types in our dataset.

	EECS / Mandarin (fine-tuned)	EECS / English (zero-shot)	Japanese (zero-shot)	Management (zero-shot)	EECS / English (fine-tuned)
PolyViLT	33.3	30.1	21.8	39.9	-
Contrastive Learning	44.7	36.0	25.6	43.6	38.9
VLM-based visual grounding (pre-trained)	8.7	10.9	11.7	10.1	-
VLM-based visual grounding (fine-tuned)	37.0	33.5	32.6	40.1	-
VLM-based AOI selection (pre-trained)	31.5	19.6	32.2	35.0	-
VLM-based AOI selection (fine-tuned)	32.2	21.1	33.1	36.1	-
AutoVAG / LLaMA 3.2 3B (ours)	**52.3**	**49.2**	**44.6**	**57.4**	**50.1**

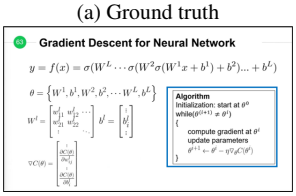
Table 2: Comparison of AutoVAG against all baselines on our in-domain and out-of-domain dataset. We adopt top-1 accuracy (%) as the evaluation metric. We fine-tune these models on the set of Mandarin-lectured EECS courses, and test their multilingual and multidisciplinary generalization on Englished-lectured EECS / Japanese-lectured courses and management courses. We then fine-tune these models on the English-lectured EECS courses to examine whether fine-tuning on a small amount of in-domain data improves performance. Note: pre-trained VLM-based methods are not fine-tuned on our dataset.



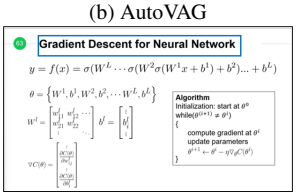
(a) Ground truth



(b) AutoVAG



(c) VLM-based visual grounding



(d) VLM-based AOI selection

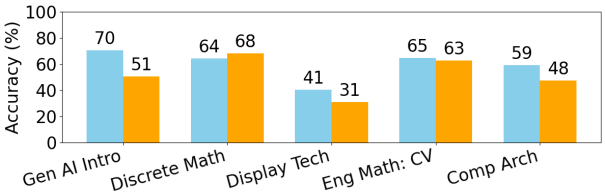
Figure 3: A visual comparison of AutoVAG against VLM-based baselines. Our method successfully detected the referred AOI, while other baselines failed.

we observed that the two VLM baselines–VLM-based visual grounding and VLM-based AOI selection–do not perform well on the dataset. Our conjecture is that the VLM backbone–Qwen 2.5 VL–is trained on natural images, and thus, it has limited capabilities in understanding the content of entire course slides, since multiple concepts are often presented concurrently in the same slide. In contrast, our method crops each slide image into AOIs and extracts language descriptions for individual AOI. It achieves much stronger performance than other baselines. The qualitative results are presented in Figure 3, where AutoVAG successfully picked out the correct AOI while others failed.

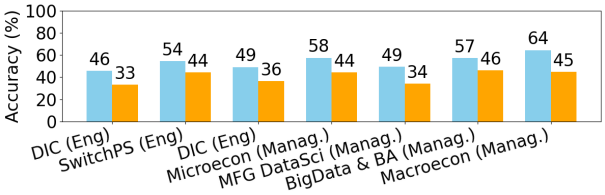
We next present more detailed comparison between our AutoVAG and contrastive learning baseline. As shown in the top figure of Figure 4, we report the accuracy on 5 of Mandarin-lectured EECS courses. Our AutoVAG outperforms the contrastive learning baseline on four of the five courses. The most notable advantage was observed in the *Introduction to Generative AI* course achieves an absolute performance gain of 19%.

Out-of-domain multilingual generalization We evaluate the multilingual generalization of our method and baselines on English-lectured EECS courses or Japanese-lectured courses. The results are presented in Table 2. Notably, our AutoVAG is able to retain its capability of pairing AOIs with subtitles in an unseen language.

For the English courses, our AutoVAG maintains 94% of the performance, while the VLM-based visual grounding and contrastive learning baseline can only retain 90% and 80% of the performance, respectively. While we observed a large performance gap between Mandarin-lectured and Japanese-lectured courses, our AutoVAG still outperforms the next best baseline—VLM-based AOI selection—by a large margin (44.6% vs. 33.1%). This performance gap is due to imprecise description of Japanese AOI content.



(a) Mandarin-lectured EECS courses



(b) English-lectured EECS courses & Management courses

Figure 4: Per-course performance on our dataset. We train our AutoVAG (cyan bars) and contrastive learning baseline (orange bars) on the set of Mandarin-lectured EECS courses. We report per-course top-1 accuracy of these methods on (a) an in-domain validation dataset and (b) two out-of-domain datasets, including English-lectured EECS courses and management courses. AutoVAG outperforms consistently.

Out-of-domain multidisciplinary generalization We evaluate the multidisciplinary generalization of our method and other baselines on management courses. As shown in Table 2, our AutoVAG outperforms all baselines by a large margin, achieving an absolute gain of 13.8%. These results demonstrate that AutoVAG is able to handle courses lectured by departments other than EECS, and verify the applicability of our method to real-world use cases. We can collect training data from departments where teachers use pointers more often, such as EECS, and deploy the system to support educational learning in other departments, where teachers seldom use pointers as visual guidance.

Likewise, we present a detailed analysis of model performance on each individual course. We report the accuracy on the three English-lectured EECS courses and the four Management courses. As shown in the bottom figure of Figure 4, our AutoVAG shows stronger multilingual and multidisciplinary generalization to unseen languages and course content, consistently across all these courses.

In-domain English-lectured EECS courses Next, we fine-tune models on the training set of English-lectured EECS courses. The performance is slightly enhanced after fine-tuning on the data. Our AutoVAG achieves an absolute performance gain of 11.2% over the contrastive learning baseline (50.1% vs. 38.9%). Notably, both methods benefit from fine-tuning on such a small amount of training data.

Ablation study on LLMs We ablate the effect of the capacity of the LLM used in AutoVAG. We conduct evaluation on the Mandarin-lectured EECS courses, comparing the performance of different-sized LLMs: LLaMa 2 7B, LLaMa 3.2 1B, LLaMa 3.2 3B, and LLaMa 3.1 8B. The results are presented in Table 3, from which we draw three conclusions.

VLM	**EECS / Mandarin (fine-tuned)**
LLaMA 2 7B	49.0
LLaMA 3.2 1B	50.1
LLaMA 3.2 3B	52.3
LLaMA 3.1 8B	**53.6**

Table 3: Ablation study on LLMs. We report the top-1 accuracy (%) of different-sized LLMs on the validation set of Mandarin courses.

First, AutoVAG can be lightweight without the need for gigantic LLMs. Using LLaMa 3.2 1B, AutoVAG already achieves 50.1% accuracy, significantly outperforming all other baselines. Second, we observe evidence of scaling laws—the performance improves with increasing model size. When using LLaMa 3.2 1B, LLaMa 3.2 3B, and LLaMa 3.1 8B, AutoVAG achieves 50.1%, 52.3%, and 53.6%, respectively. Lastly, the performance improves with newer LLMs. With similar-capacity LLMs (7/8B), AutoVAG achieves better performance when using LLaMa 3.1 than LLaMa 2. Meanwhile, our modularized design allows us to improve AutoVAG by plugging in newer LLMs. These results demonstrate the maintainability of our method.

Method	100 %	50 %	25 %
Contrastive Learning	44.7	38.8 (-5.9)	35.6 (-9.1)
AutoVAG / LLaMA 3.2 3B (ours)	52.3	50.6 (-1.7)	50.2 (-2.1)

Table 4: Ablation study on data efficiency. We report the top-1 accuracy (%) of AutoVAG and contrastive learning baseline , trained with varying-sized dataset, on Mandarin-lectured courses.

Analysis on data efficiency We present a deeper investigation on the effect of dataset size on our AutoVAG and the contrastive learning baseline. We compare the performance of these models trained on 25%, 50%, and 100% of the original training set of Mandarin-lectured EECS courses. As shown in Table 4, the performance of the contrastive learning baseline drops significantly when less training data is available. In contrast, our method is able to retain 96% of the performance using 25% of the original training data. Our conjecture is that, with rich commonsense knowledge encoded in the LLMs, models that build upon these LLMs are able to handle visual-language tasks more efficiently.

Explainability via reasoning A key advantage of AutoVAG is the interpretability of its choice. As described earlier, we developed a fine-tuned variant of AutoVAG capable of generating rationales for its selections. Figure 5 shows an example of this variant in use: when presented with subtitle text discussing the weight matrix and bias vector in a neural network, AutoVAG correctly selects AOI 4 and provides an accompanying rationale for its choice. This transparency is achieved with only a minor and acceptable trade-off in accuracy (a decrease from 52.3% to 49.1%) and is essential for fostering user trust in educational applications.

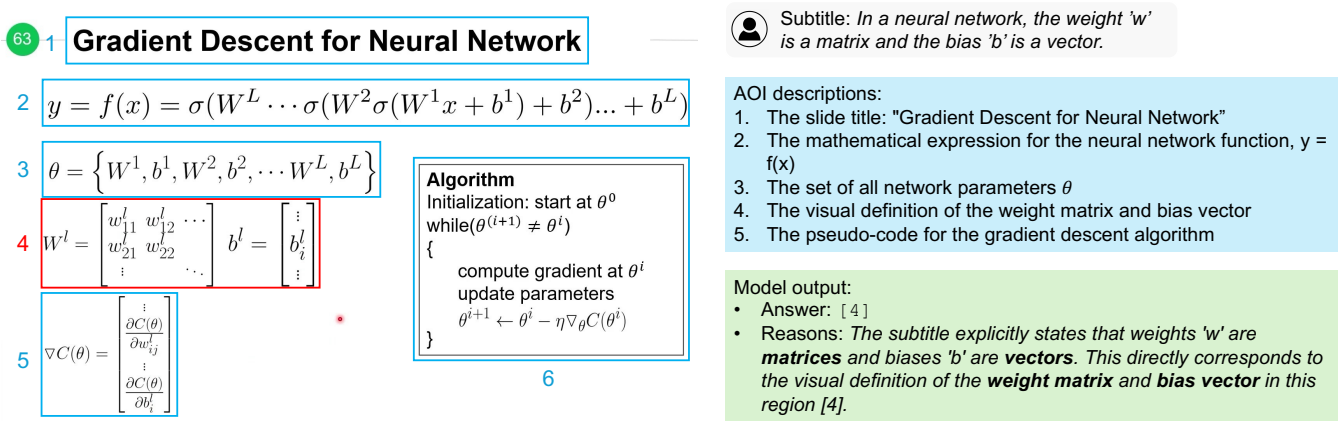
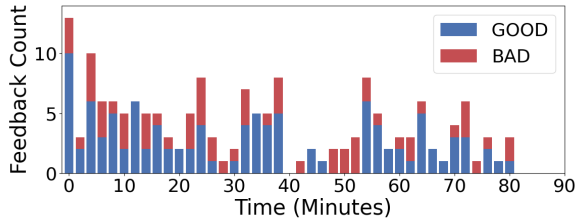
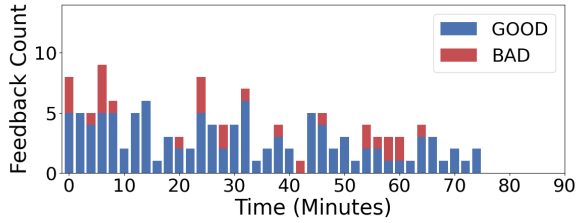


Figure 5: An example of our framework's interpretability. Given the subtitle context and several AOI options from the slide on the left, the model not only selects the correct AOI [4] but also generates a human-readable rationale explaining its reasoning.



(a) Constrative learning



(b) AutoVAG

Figure 6: Temporal distribution of user feedback. We present real-time feedback provided by the users.

User Study

To qualitatively demonstrate the efficacy of our AutoVAG, we conducted a small-scale user study involving seven undergraduate students from different universities. Participants were instructed to watch a 90-minute lecture video on an unfamiliar computer science topic. The video was preprocessed to visually emphasize the AOIs predicted by either the contrastive learning baseline or AutoVAG. The participants were evenly divided into two groups, with each group viewing the video processed by a different method.

While watching the video, participants were instructed to provide real-time feedback by clicking either the GOOD or BAD button. A GOOD response indicated that the participant found the visual aid helpful and the annotation accurate, whereas a BAD response indicated the opposite. Upon completing the task, each participant also filled out a questionnaire. See the Appendix for details.

The results indicate that users perceived the annotations generated by AutoVAG as significantly better than those produced by the contrastive learning baseline. On average, AutoVAG resulted in 9.0 BAD clicks per user, compared to 15.3 BAD clicks per user for the baseline. Figure 6 shows the temporal distribution of GOOD and BAD clicks for both methods. Notably, with AutoVAG, users left no BAD clicks for 51% of the video duration, whereas with the contrastive learning baseline, BAD clicks were observed throughout the entire video. Finally, questionnaire responses were largely positive toward AutoVAG, with participants frequently describing its annotations as *very good*, while criticizing those from the contrastive learning baseline as *counterproductive*.

Conclusion

We have presented a scalable framework for grounding spoken lecture content to visual AOIs on slides, a capability critical for focusing student attention in video-based learning. Our approach rests on two core innovations: (1) an automated, weakly-supervised annotation pipeline that generates subtitle-AOI pairs with minimal human effort, and (2) a text-to-text alignment strategy where VLMs produce rich AOI descriptions and LLMs perform multiple-choice matching.

Extensive experiments on 37 undergraduate EECS courses demonstrate the superior performance of our AutoVAG, outperforming strong baselines such as PolyVilT, contrastive learning, VLM-based visual grounding and VLM-based AOI selection by a large margin. Our method also shows strong multilingual and multidisciplinary generalization on out-of-domain datasets. Furthermore, we highlight practical benefits including model-size flexibility and data efficiency in low-resource settings. Our framework unlocks LLMs' advanced semantic reasoning for precise subtitle-AOI alignment in educational learning.

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are included} {(yes/partial/no)}
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```

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3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **partial**

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) **yes**

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) **yes**

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) **No**

If yes, please address the following points:

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4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) **NA**

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) **NA**

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) **NA**

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4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) **NA**

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4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) **NA**

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) **NA**

Implementation Details and Hyperparameters

This section provides the detailed configurations and hyperparameters used for training the main models and key baselines discussed in this paper, ensuring the reproducibility of our experiments.

LLM Fine-tuning (Our Method) Our primary models, the LLaMA variants, were fine-tuned using the following configuration:

- **Fine-tuning Method:** Low-Rank Adaptation (LoRA)
- **LoRA Rank:** 32
- **LoRA Alpha:** 16
- **LoRA Dropout:** 0.05
- **Learning Rate:** 2e-5
- **Epochs:** 3
- **Optimizer:** AdamW
- **Scheduler:** Cosine

Advanced VLM Baselines Fine-tuning (Qwen-2.5 VL) The VLM-based baselines (Direct Grounding and Referring Selection) were fine-tuned using the Qwen-2.5 VL model with the following parameters:

- **Fine-tuning Method:** LoRA
- **LoRA Rank:** 32
- **LoRA Alpha:** 16
- **LoRA Dropout:** 0.05
- **Learning Rate:** 2e-4
- **Epochs:** 3
- **Image Max Pixels:** 1280 * 28 * 28
- **Image Min Pixels:** 256 * 28 * 28
- **Optimizer:** AdamW
- **Scheduler:** Cosine

Contrastive Learning Fine-tuning (Baselines) We trained two variants of contrastive learning models, both using the Chinese-CLIP encoder as the backbone. The text-to-text baseline (used in the main paper) and the classic text-to-image baseline (used in the ablation study) were trained with the following consistent hyperparameters:

- **Learning Rate:** 1e-4
- **Epochs:** 100
- **Optimizer:** AdamW
- **Scheduler:** Linear

Phi-4 Multimodal Model Fine-tuning (for VLM Descriptions) The Phi-4 Multimodal model was fine-tuned to generate structured descriptions for Chinese-language AOIs. Our preliminary analysis showed that the model's vision component could already understand Chinese characters, so the primary goal was instruction tuning for output formatting.

- **Training Data:** A set of 2,243 unique slide pages (containing 9,726 AOIs) from public YouTube courses, with ground-truth descriptions generated by GPT-4o.
- **Training Method:** Instruction tuning with LoRA, following the official vision fine-tuning example for `microsoft/Phi-4-multimodal-instruct`.
- **Key Hyperparameters:**
  - **Epochs:** 1
  - **Learning Rate:** 4.0e-5
  - **Optimizer:** AdamW
  - **Scheduler:** Linear

Ablation and Additional Baseline Studies

To further validate our design choices and the robustness of our primary method, we conducted several additional experiments.

Ablation on Retrieval Paradigm: Image-based MCQA

To investigate the importance of our "vision-to-text" VLM description stage, we experimented with a variant where the LLM is presented with visual options instead of textual ones. In this setup, the AOI regions are provided to a multimodal LLM directly as image patches, and the model must select the most relevant image patch for the given subtitle. As shown in Table 5, the Top-1 accuracy for this image-based MCQA model was only **25.7%** on the Mandarin validation set, which is substantially lower than our main model's 52.3%. This result strongly suggests that converting visual content into rich, descriptive text is a critical step that allows the LLM to better leverage its reasoning capabilities.

Analysis of a Classic Text-to-Image CLIP Baseline

To directly quantify the benefit of our text-centric pipeline, we evaluated a baseline that uses a classic cross-modal (text-to-image) contrastive learning approach. For this experiment, we used the same **Chinese-CLIP** model from our main contrastive baseline, but in its intended multimodal configuration. The model was trained to directly match the spoken **subtitle embeddings** with the embeddings of the raw **AOI images**, without the intermediate VLM description step. The performance of this direct text-to-image model on the Mandarin validation set was **33.1%**, as reported in Table 5. This result is significantly lower than both our text-to-text Contrastive Learning baseline (44.7%) and our main

AutoVAG model (52.3%), strongly validating our hypothesis that transforming visual information into text before retrieval is a more effective paradigm for this task.

Summary of Ablation Studies The results from our ablation studies, summarized in Table 5, provide a clear conclusion. Both methods that bypass our VLM-to-text pipeline—the direct Text-to-Image CLIP model and the Image-based MCQA model—perform significantly worse than our text-centric approaches across all evaluation scenarios. This demonstrates that the intermediate step of converting complex visual AOIs into rich textual descriptions is not just a design choice, but a critical component for achieving high performance in this domain.

User Study Materials

This section provides the full questionnaire administered to participants in our pilot user study and an illustration of the user interface used to collect real-time feedback, as referenced in the main paper.

Learning Performance Test

The following questions were used to assess the learning performance of participants.

1. A B-tree of minimum degree $t = 4$ and of height $h = 3$ will have a maximum of _____ keys.

2. Explain when and why the root of a B-tree is allowed to contain fewer than $t - 1$ keys.

3. Perform the following operations on the B-tree in Figure 7 in the given order. After each operation, draw the resulting B-tree.

(a) Insert 115

(b) Insert 21

(c) Insert 200

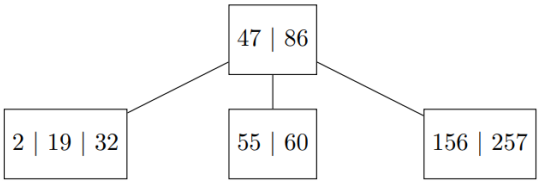

Figure 7: The B-tree used for question 1.3 of the learning performance test.

Subjective Experience Questionnaire

After interacting with the system, participants were asked to complete the following questionnaire to gather their subjective feedback on the helpfulness and usability of the automated highlights.

Five-point Likert Scale Questions (1=Strongly Disagree, 5=Strongly Agree)

1. Do you think the highlights provided by the system were accurate and helpful?

2. Could you clearly understand what the system was highlighting in the video?

3. Do you think the highlights helped you focus on the key points the instructor was explaining?

4. Did the system's highlights cause additional distraction while watching the video?

5. If this system were officially launched, would you be willing to use it again?

Open-ended Questions

1. What shortcomings did you find in the system? Are there any areas for improvement?

2. Please use one sentence to describe your overall feeling about this automated highlighting system.

User Interface for Real-time Feedback

During the user study, we provided participants with a simple interface to give real-time feedback on the quality of the automated highlights. The interface featured "Good" and "Bad" buttons below the video player, allowing users to quickly indicate whether they found the current highlight to be relevant and accurate. A screenshot of this interface is shown in Figure 8.

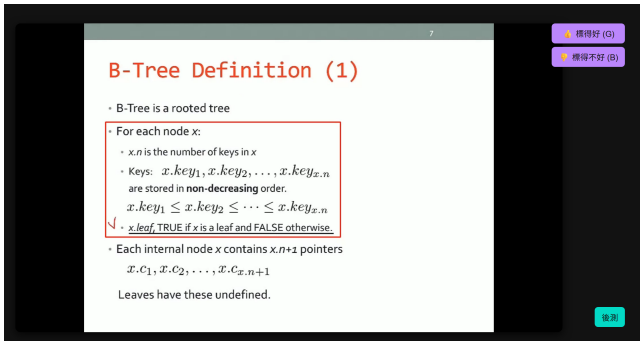

Figure 8: Screenshot of the video player interface shown to participants for providing real-time feedback on the automated highlights using "Good" and "Bad" buttons.

Table 5: Top-1 Accuracy (%) comparison of our main text-centric methods against ablation baselines that bypass the VLM-to-text description stage. The significant performance drop in the ablation models validates the effectiveness of our text-centric paradigm.

Model / Paradigm	**EECS / Mandarin**	**EECS / English**	**Japanese**	**Management**
Ablation Baselines (Bypassing the Text-Centric Pipeline)				
Image-based MCQA	25.7	20.0	26.4	27.6
Text-to-Image CLIP	33.1	28.4	18.4	36.9
Our Text-Centric Methods (from Main Paper)				
Contrastive Learning (Text-to-Text)	44.7	36.0	25.6	43.6
AutoVAG (Ours)	**52.3**	**49.2**	**44.6**	**57.4**